Yike (Echo) Li

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Education

Swarthmore College — Swarthmore, PA
B.A. in Mathematics and Linguistics — *Expected May 2028*

Technical Skills

Programming: Python, SQL, C++, Java, HTML, R
AI/NLP: NumPy, pandas, scikit-learn, PyTorch, LLM APIs & Integration, Prompt Engineering, AI Agent Workflows
Formalisms: Lean, PDDL
Languages: Mandarin Chinese, English, Japanese

Experience

Research Intern — Oct. 2026 – Present
Haz Lab — Advisor: Prof. Harry Zhang — *Philadelphia, PA*

- Investigating LLM generalization in planning-problem formalization by inducing reusable generator programs for regular cases while explicitly representing exceptions to avoid spurious generalizations.
- Developing a controlled benchmark across four UNRAVEL planning domains, injecting exceptions at varying rates and presenting instances in multiple formats, including flat text, tables, and unstructured prose.
- Designing probing experiments across multiple LLMs and reasoning-effort levels to evaluate pattern induction, exception handling, and robustness to input representation.

Undergraduate Researcher (NLP) — Jul. 2026 – Present
Bryn Mawr College — Research with Prof. Adam Poliak — *Bryn Mawr, PA*

- Co-authored **FINNLI-PARA**, a study of paraphrase robustness in financial NLI; validated lexical, syntactic, and plain-language rewrites for semantic equivalence and NLI-label preservation.
- Evaluating prediction consistency under meaning-preserving rewrites and developing **FORECAST**, a benchmark distinguishing predictions, intentions, goals, and possibilities in corporate disclosures.

Undergraduate Research Assistant — Sep. 2026 – Present
Swarthmore College — Research with Prof. Jonathan Washington — *Swarthmore, PA*

- Evaluated a rule-based Turkish morphological analyzer (Apertium/HFST) on 4.3M+ tokens from the OPUS SETimes corpus, achieving 95.0% token-level coverage and establishing a baseline for morphological analysis; Conducted large-scale error analysis on 216K+ unrecognized tokens, identifying systematic issues involving proper-name suffixes, apostrophe handling, and morphological segmentation.
- Building a small Turkish GPT-style language model based on TurEngMix benchmark and implementing controlled experiments on linguistic structure and data availability in low-resource Hybrid MT.

Digital Scholarship Intern — Sep. 2026 – Present
Swarthmore College Libraries — *Swarthmore, PA*

- Developing HTML/CSS workflows for digital archives and metadata-based collections; organizing front-end development workshops with the college's Center of Innovation.

Projects

Kickapoo–English Rule-Based Machine Translation | *Python, Apertium, Shell, Git* — Feb. 2026 – May 2026

- Built an Apertium-based Kickapoo–English MT system with finite-state morphology, lexical and structural transfer; expanded monolingual and bilingual dictionaries by **132 stems**.
- Developed Python/Shell evaluation pipelines for transducer coverage, ambiguity, WER/PER, and unknown-word rates; implemented Constraint Grammar rules for disambiguation.

Selected Publication

Yike Li, Adam Poliak. "Same Meaning, Different Wording: Evaluating Paraphrase Robustness in Financial Natural Language Inference." *FinNLP 2026 (EMNLP Workshop).*